

A GRADE 10 ENGLISH EXPLORATION PROJECT

BEYOND THE HORIZON.

Exploring the unknown.

Five frontiers. Extraordinary discoveries.
One shared future.

SPACE / DEEP OCEAN / POLAR REGIONS / RAINFORESTS / MOUNTAINS



What is exploration?

A journey with a question.

Investigating places and ideas to learn something new.

01

Ask a question

What do we want to understand?

02

Find evidence

Observe, measure, and record.

03

Share knowledge

Help others learn from the results.

Tourism is mainly for enjoyment. Scientific exploration has a research purpose. A place can be new to a researcher and already familiar to local people.

Sources: NOAA / Smithsonian Tropical Research Institute

Why do humans explore?

Curiosity turns into knowledge.

01

Curiosity

Understand Earth, other worlds, and the life around us.

02

Better solutions

Develop technology and answer useful questions.

03

A shared future

Use discoveries to understand and protect nature.

OUR PRESENTATION FOCUS

Five environments / Benefits and risks / Problems and solutions

Sources: NASA Science / NOAA / Smithsonian Tropical Research Institute

Space exploration

The Moon and Mars



THE MOON & MARS

Apollo 17 lunar rover / NASA, 1972

ADVANTAGES

- Understand planets and the universe
- Develop robots, sensors, and technology
- Study conditions for life

DISADVANTAGES

- Enormous costs and long preparation
- Harmful radiation
- Dangerous missions and equipment failures

A POSSIBLE SOLUTION

Use radiation monitors, storm shelters, and tested shielding. These reduce exposure but cannot remove every risk.

Deep ocean exploration

A world beneath the surface



THE DEEP OCEAN / MARIANA TRENCH

Atolla jellyfish, 880 m deep / NOAA

ADVANTAGES

- Discover species and ecosystems
- Understand Earth's climate
- Map habitats for marine protection

DISADVANTAGES

- Crushing water pressure
- Expensive ships and equipment
- Damage to fragile habitats

A POSSIBLE SOLUTION

Use pressure-tested housings and remotely operated vehicles, or ROVs, controlled from a ship.

Polar exploration

Antarctica and the Arctic



ANTARCTICA & THE ARCTIC

Antarctic icebergs / Anne Dirkse

ADVANTAGES

- Learn about past climate from ice
- Study ice and polar wildlife
- Build international cooperation

DISADVANTAGES

- Extreme cold, storms, and ice cracks
- Isolation and difficult rescue
- Fuel spills and disturbed wildlife

A POSSIBLE SOLUTION

Use insulated clothing and shelters, check weather forecasts, and postpone travel when conditions are dangerous.

Rainforest exploration

Understanding the living world



TROPICAL FORESTS / THE AMAZON

Tropical canopy, Queensland / CSIRO

ADVANTAGES

- Discover species and biodiversity
- Understand forests and carbon storage
- Research plants with medical potential

DISADVANTAGES

- Disease, insects, and difficult terrain
- Heat and humidity
- Damage to habitats and communities

A POSSIBLE SOLUTION

Plan health protection with experts, carry safe water and first aid, and work with trained local guides.

Sources: Smithsonian Tropical Research Institute / Smithsonian ForestGEO / Royal Botanic Gardens, Kew

Mountain exploration

The Himalayas and beyond



THE HIMALAYAS & OTHER RANGES

Himalayan peaks from Gokyo Ri / Gavin Challand

ADVANTAGES

- Improve maps and geographical knowledge
- Study glaciers, weather, and ecosystems
- Gain skills and personal achievement

DISADVANTAGES

- Dangerous altitude sickness
- Avalanches, storms, and falls
- Pollution and abandoned equipment

A POSSIBLE SOLUTION

Allow time to adjust through gradual ascent. If illness develops, stop ascending, seek expert help, and descend when needed.

Benefits and risks

Every frontier has a trade-off.

WHAT WE GAIN

- New discoveries and useful knowledge
- Better tools and technology
- A clearer picture of our changing planet

WHAT WE RISK

- Large costs and competing needs
- Danger to explorers and support teams
- Damage to habitats and communities

A useful discovery must justify its cost, danger, and impact.

Sources: NASA / NOAA Ocean Exploration / British Antarctic Survey / US National Park Service

Common problems

Different places. Shared challenges.

01

Extreme conditions

Cold, pressure, radiation, or altitude: protection and tested equipment.

02

Limited supplies

Food, water, power, or spare parts: careful planning and backups.

03

Communication

Distance and weak signals: reliable systems and clear procedures.

04

Help far away

Delayed rescue: training, emergency plans, and safe stopping rules.

Sources: NASA / NOAA Ocean Exploration / US National Park Service

Technology and solutions

Explore smarter. Work together.

01

Robotic explorers

Rovers and ROVs keep people away from the most dangerous conditions.

02

Maps and remote sensing

Satellites, sonar, and cameras help teams plan and observe.

03

Protection and planning

Protective equipment, backups, and forecasts reduce risks.

04

Shared knowledge

Scientists, local guides, and communities learn together.

Sources: NASA Science / NOAA Ocean Exploration / British Antarctic Survey

The environmental impact

Discovery should not mean damage.

01

Observe first

Use cameras and remote tools where possible. Limit sampling.

02

Reduce the impact

Remove waste, prevent spills, and avoid disturbing wildlife.

03

Respect people

Follow local rules, seek agreement, and share benefits.

Research can guide conservation. Careless exploration can damage what it studies.

Should we keep exploring?

A question worth discussing.

01

Yes, because...

Discoveries can improve technology, explain our world, and support conservation.

02

But we must ask...

Could the money meet urgent needs? Can we manage the danger and environmental cost?

Our answer: yes - with a clear purpose, careful planning, and respect.

Our conclusion

Keep the curiosity. Carry the responsibility.

We believe that exploration is important because it helps humanity discover new things, develop technology, and understand our planet. However, exploration must be responsible, safe, and respectful of the environment.

STAY CURIOUS / STAY SAFE / SHOW RESPECT

Thank you.

What would you explore next?

Which frontier would you choose?

How would you explore it responsibly?

SPACE / DEEP OCEAN / POLAR REGIONS / RAINFORESTS / MOUNTAINS



RESEARCH AND REFERENCES

Curiosity, backed by science.

Sources checked 8 October 2026. Comparison ratings and final opinion are classroom judgments.

NASA: Apollo 11

The 1969 Moon landing, crew, and lunar research.

<https://www.nasa.gov/mission/apollo-11/>

NASA: Space radiation and human exploration

Radiation hazards and protective technology; shielding has limits.

<https://www.nasa.gov/humans-in-space/space-radiation-wont-stop-nasas-human-exploration/>

NOAA: What is a hydrothermal vent?

The discovery of vent ecosystems in 1977 and life supported by chemical energy.

<https://oceanservice.noaa.gov/facts/vents.html/volcanoes.html>

NOAA: Journey to Earth's largest habitat

Ocean exploration and understanding the climate.

<https://www.noaa.gov/stories/story-map-explore>

NASA Science: Perseverance exploring the front of the delta

The 2021 landing and evidence of an ancient river and lake at Jezero.

<https://science.nasa.gov/resource/perseverance-exploring-the-front-of-the-delta/>

NASA Science: Moon facts

The Moon's very thin atmosphere, called an exosphere.

<https://science.nasa.gov/moon/facts/>

NOAA Ocean Exploration: How robots uncover the mysteries of the deep

ROVs, pressure-resistant housings, deep-ocean darkness, and communication.

<https://oceanexplorer.noaa.gov/explainers/technology/>

British Antarctic Survey: Ice cores and climate change

Ancient air bubbles in ice and records of past climate.

https://legacy.bas.ac.uk/bas_research/science_briefings/icecorebriefing.php

RESEARCH AND REFERENCES

Curiosity, backed by science.

Sources checked 8 October 2026. Comparison ratings and final opinion are classroom judgments.

British Antarctic Survey: Environmental protection

Environmental planning, protection of wildlife, and the Antarctic Treaty System.

<https://www.bas.ac.uk/about/where-we-work/environmental-protection/>

Smithsonian Tropical Research Institute: Biodiversity

Species discovery and the scientific study of tropical biodiversity.

<https://stri.si.edu/discipline/biodiversity>

Royal Botanic Gardens, Kew: Into the wild: plant drug discovery

Plant research, traditional knowledge, and fieldwork in remote forests.

<https://www.kew.org/read-and-watch/plant-medicine-drug-discovery>

Royal Geographical Society: Everest 1953

The first confirmed Everest summit by Tenzing Norgay and Edmund Hillary.

<https://www.rgs.org/our-collections/buy-and-license-images/platinum-prints/everest-1953>

Norwegian Polar Institute: The South Pole expedition

Amundsen's team reached the South Pole on 14 December 1911.

<https://nettarkiv.npolar.no/sorpolen2011.npolar.no/en/diary/south-pole/2011-10-19-the-day-we-should-have-set-off.html>

Smithsonian ForestGEO: Climate and 35 years of forest observations

The 1982 tree census and long-term research at Barro Colorado Island, Panama.

<https://forestgeo.si.edu/demographic-trends-and-climate-over-35-years-barro-colorado-50-ha-plot>

Royal Botanic Gardens, Kew: Spatial Analysis and Data Science

Remote sensing and spatial data support forest and biodiversity research.

<https://www.kew.org/science/our-science/departments/ecosystem-stewardship/spatial-analysis-and-data-science>

US National Park Service: Mountaineering medical issues

Altitude illness, gradual ascent, cold injuries, and the importance of descent.

<https://www.nps.gov/dena/planyourvisit/part2medicalissues.htm>

Real places. Real photographs.

Hello, World - Earth from Artemis II, April 2, 2026

NASA / Reid Wiseman / Artemis II

<https://www.nasa.gov/image-article/hello-world/>

Public domain (NASA astronaut photograph) / <https://www.nasa.gov/nasa-brand-center/images-and-media/>

Apollo 17 lunar rover in the Taurus-Littrow Valley

NASA/GSFC/Arizona State University

<https://science.nasa.gov/resource/apollo-17-landing-site-the-taurus-littrow-valley/>

Public domain (NASA Apollo photograph) / <https://www.nasa.gov/nasa-brand-center/images-and-media/>

Atolla jellyfish at 880 meters depth, Windows to the Deep 2018

NOAA Ocean Exploration, Windows to the Deep 2018

<https://oceanexplorer.noaa.gov/multimedia/daily-image-media-20210319/>

Public domain; preserve the listed NOAA image credit / <https://oceanexplorer.noaa.gov/faqs/>

Blue Ice in Antarctica, Livingston Island, November 2012

Anne Dirkse (www.annedirkse.com), via Wikimedia Commons

https://commons.wikimedia.org/wiki/File:Blue_Ice_in_Antarctica.jpg

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Rainforest canopy, Bellenden Ker, North Queensland, November 1989

Robert Kerton / CSIRO, via Wikimedia Commons

https://commons.wikimedia.org/wiki/File:CSIRO_ScienceImage_1336_Rainforest_Canopy.jpg

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Everest, Lhotse and Makalu from Gokyo Ri, April 2006

Gavin Challand (Gavidipus), via Wikimedia Commons

https://commons.wikimedia.org/wiki/File:Everest_Lhotse_and_Makalu_from_Gokyo_Ri.jpg

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The rainforest is Queensland. The Atolla jellyfish was illuminated by an underwater robot.